
3AL Tutorial 3 2024 Solutions

1. Prove that G/G is isomorphic to the trivial group.

Note that G only has one coset in G since $aG = G$ for all $g \in G$. Thus G/G is a one element group, so it must be the trivial group.

Note that we cannot use Lagrange's Theorem to say $|G/G| = |G|/|G| = 1$ because that only works for finite groups G .

2. Prove, in two separate ways, that $G/\{1\} \cong G$ (with and without using the First Isomorphism Theorem).

First, we prove directly without using the First Isomorphism Theorem. So, we construct an isomorphism $\varphi : G/\{1\} \rightarrow G$. If we let H denote $\{1\}$, then the elements of $G/\{1\} = G/H$ are the cosets gH for each $g \in G$. So, let's define $\varphi(gH) = g$. Notice that $gH = g\{1\} = \{g\}$ and so $gH = g'H$ implies $g = g'$, thus $\varphi(gH) = \varphi(g'H)$. Therefore, φ is well-defined. Moreover, it is injective since if $gH = g'H$, then $\{g\} = \{g'\}$ and $g = g'$ so $\varphi(gH) = \varphi(g'H)$. Finally, it is surjective because each element $g \in G$ has $\varphi(gH) = g$.

To use the First Isomorphism Theorem, we construct an epimorphism $\varphi : G \rightarrow G$ with kernel $\{1\}$. The most natural epimorphism would be the map that takes an element g to itself. So, let's define $\varphi(g) = g$ and check if it works. It is clear that φ preserves the operations and is surjective. Now we show that $\ker \varphi = \{1\}$. Obviously $1 \in \ker \varphi$, but suppose that there is some $g \neq 1$ in $\ker \varphi$. Then $\varphi(g) = 1$, but by the definition of φ , we also have $\varphi(g) = g$. Thus, $g = 1$. Therefore $\ker \varphi = \{1\}$. Finally, by the First Isomorphism Theorem, $G/\{1\} \cong G$.

3. Let $T_2(\mathbb{R})$ be the multiplicative group of 2×2 upper triangular matrices with entries from $\mathbb{R}$ with nonzero determinant. Consider the following subset of $T_2(\mathbb{R})$:

$$U = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} : b \in \mathbb{R} \right\}.$$

- (a) Prove that U is a normal subgroup of $T_2(\mathbb{R})$.
- (b) Show that $U \cong \mathbb{R}$ (where $\mathbb{R}$ is the additive group of real numbers).
- (c) Use the First Isomorphism Theorem to show that $T_2(\mathbb{R})/U \cong \mathbb{R}^\times \times \mathbb{R}^\times$, where $\mathbb{R}^\times$ is the multiplicative group of nonzero real numbers.

(a) First, notice that the identity matrix is in U (take $b = 0$). Now we show that U is closed under the operation and taking inverses. Take $\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix} \in U$. Then,

$$\begin{aligned} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix}^{-1} &= \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -b' \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & b - b' \\ 0 & 1 \end{pmatrix} \in U. \end{aligned}$$

Therefore, U is a subgroup of $T_2(\mathbb{R})$. To prove that U is normal in $T_2(\mathbb{R})$, we will show that $AUA^{-1} \subseteq U$ for all $A \in T_2(\mathbb{R})$. Let $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in T_2(\mathbb{R})$ and $\begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix} \in U$. Then

$$\begin{aligned} &\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{ac} & \frac{-b}{ac} \\ 0 & a \end{pmatrix} \\ &= \frac{1}{ac} \begin{pmatrix} a & ab' + b \\ 0 & c \end{pmatrix} \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} \\ &= \frac{1}{ac} \begin{pmatrix} ac & a^2b' \\ 0 & ca \end{pmatrix} \\ &= \begin{pmatrix} 1 & \frac{ab'}{c} \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Therefore, $AUA^{-1} \subseteq U$ for all $A \in T_2(\mathbb{R})$. So $U \triangleleft T_2(\mathbb{R})$. (Note that $c \neq 0$ because otherwise the determinant of our matrix in $T_2(\mathbb{R})$ would be 0, but we specified that $T_2(\mathbb{R})$ only has matrices with nonzero determinant).

(b) We define an isomorphism $\varphi : U \rightarrow \mathbb{R}$ as follows:

$$\varphi \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} = b.$$

Note that

$$\varphi \left(\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix} \right) = \varphi \begin{pmatrix} 1 & b+b' \\ 0 & 1 \end{pmatrix} = b+b' = \varphi \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} + \varphi \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix}.$$

So φ preserves operations. Suppose that

$$\varphi \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} = \varphi \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix}.$$

Then $b = b'$ and so

$$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix}.$$

Thus φ is injective. Moreover, for each $x \in \mathbb{R}$, we have $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \in U$ such that

$$\varphi \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = x.$$

Hence φ is surjective. Finally, we conclude that φ is an isomorphism, so $U \cong \mathbb{R}$.

(c) We need to construct an epimorphism $\varphi : T_2(\mathbb{R}) \rightarrow \mathbb{R}^\times \times \mathbb{R}^\times$ with kernel U .

Elements of $T_2(\mathbb{R})$ have the form $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$. So, we define

$$\varphi \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = (a, c).$$

Notice that

$$\varphi \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} a' & b' \\ 0 & c' \end{pmatrix} \right) = \varphi \begin{pmatrix} aa' & bc' + b'c \\ 0 & cc' \end{pmatrix} = (aa', cc'),$$

and

$$\varphi \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \varphi \begin{pmatrix} a' & b' \\ 0 & c' \end{pmatrix} = (a, c)(a', c') = (aa', cc').$$

So φ preserves operations. For all $(x, y) \in \mathbb{R}^\times \times \mathbb{R}^\times$, we have $\begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix} \in T_2(\mathbb{R})$ and $\varphi \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix} = (x, y)$. Thus φ is an epimorphism. The identity of $\mathbb{R}^\times \times \mathbb{R}^\times$ is $(1, 1)$. Thus the kernel of φ consists of matrices in $T_2(\mathbb{R})$ where $a = c = 1$. That is, $\ker \varphi = U$. By the First Isomorphism Theorem, $T_2(\mathbb{R})/U \cong \mathbb{R}^\times \times \mathbb{R}^\times$.

4. The unit circle in the complex plane can be expressed as the set

$$S^1 = \{e^{2i\pi\theta} : \theta \in \mathbb{R}\}.$$

- (a) Prove that S^1 is a subgroup of $\mathbb{C}^\times$.
- (b) Consider the additive group $\mathbb{R}$ with subgroup $\mathbb{Z}$. What do the elements of the factor group $\mathbb{R}/\mathbb{Z}$ look like? Give (at least) two examples.
- (c) Prove that $\mathbb{R}/\mathbb{Z} \cong S^1$.

HINT: Recall that each $z \in \mathbb{C}$ can be written as $z = re^{i\theta}$ for some $r, \theta \in \mathbb{R}$. What does the product of two complex numbers in polar form look like? What is the multiplicative identity?

(a) Notice that S^1 is nonempty, it contains the identity of $\mathbb{C}^\times$ since $1 = e^{2i\pi \times 0} = e^0$. Let $e^{2i\pi\theta}, e^{2i\pi\omega} \in S^1$. Then

$$e^{2i\pi\theta} (e^{2i\pi\omega})^{-1} = e^{2i\pi\theta - 2i\pi\omega} = e^{2i\pi(\theta - \omega)} \in S^1$$

So S^1 is a subgroup of $\mathbb{C}^\times$ under multiplication.

(b) For example, $\pi + \mathbb{Z} = \{\pi, \pi + 1, \pi - 1, \dots\}$ and $0.1 + \mathbb{Z} = \{0.1, 1.1, -0.9, 2.1, \dots\}$. Notice that each element of these cosets consists of real numbers that differ by an integer.

(c) For this, we will use the First Isomorphism Theorem. We'll start by defining an epimorphism $\varphi : \mathbb{R} \rightarrow S^1$. Because each element of S^1 can be expressed in the form $e^{2i\pi\theta}$, for some real number θ , it makes sense to define the map by $\varphi(x) = e^{2i\pi x}$. We

need to show that this is an epimorphism. Let $x, y \in \mathbb{R}$, then

$$\varphi(x)\varphi(y) = e^{2i\pi x}e^{2i\pi y} = e^{2i\pi(x+y)} = \varphi(x+y).$$

(Recall that we are considering $\mathbb{R}$ as an additive group) It is clear, from how S^1 is defined, that φ is surjective. Now we need to find the kernel of φ . The identity of S^1 is 1. Now, $e^{2i\pi x} = 1$ if and only if x is an integer (think of this graphically, we are doing a rotation of a multiple of 2π radians). So, $x \in \ker \varphi$ if and only if $x \in \mathbb{Z}$. That is, $\ker \varphi = \mathbb{Z}$. So, the First Isomorphism Theorem says that $\mathbb{R}/\mathbb{Z} \cong S^1$.

5. Prove that $\mathbb{R}^\times/\{1, -1\} \cong \mathbb{R}^{>0}$, where $\mathbb{R}^{>0}$ is the multiplicative group of positive real numbers.

We define an epimorphism $\varphi : \mathbb{R}^\times \rightarrow \mathbb{R}^{>0}$ by $\varphi(x) = x^2$ (you could use $|x|$ instead). Note that for $x, y \in \mathbb{R}^\times$, we have $\varphi(xy) = (xy)^2 = x^2y^2 = \varphi(x)\varphi(y)$, so φ preserves operations. Moreover, φ is surjective because for all $x \in \mathbb{R}^{>0}$, we have $\sqrt{x} \in \mathbb{R}^\times$ and $\varphi(\sqrt{x}) = x$.

Notice that $x \in \ker \varphi$ if and only if $\varphi(x) = x^2 = 1$. That is, $x = 1$ or -1 . So, $\ker \varphi = \{1, -1\}$. Then by the First Isomorphism Theorem, $\mathbb{R}^\times/\{1, -1\} \cong \mathbb{R}^{>0}$.

6. Prove that $\mathbb{Z}_{18}/\langle [3]_{18} \rangle \cong \mathbb{Z}_3$. (The subscript 18 denotes that the equivalence class is using modulo 18)

We can construct an epimorphism $\varphi : \mathbb{Z}_{18} \rightarrow \mathbb{Z}_3$ with kernel $\langle [3]_{18} \rangle$. If we just focus on constructing a map between the two groups, the most natural idea would be the mapping $[n]_{18} \mapsto [n]_3$. The map is well-defined since if $n \equiv m \pmod{18}$, then $n \equiv m \pmod{3}$ since 3 is a divisor of 18 (i.e. $18 \mid (n - m)$ implies $3 \mid (n - m)$). It is a homomorphism since, for $[n]_{18}, [m]_{18} \in \mathbb{Z}_{18}$, we have, by the properties of modular arithmetic,

$$\varphi([n]_{18} + [m]_{18}) = \varphi([n + m]_{18}) = [n + m]_3 = [n]_3 + [m]_3 = \varphi([n]_{18}) + \varphi([m]_{18}).$$

Moreover, φ is an epimorphism since $\varphi([0]_{18}) = [0]_3$, $\varphi([1]_{18}) = [1]_3$ and $\varphi([2]_{18}) = [2]_3$. Now let's check the kernel of the map. Hopefully it will be $\langle [3]_{18} \rangle$. Notice that

$$\langle [3]_{18} \rangle = \{0, 3, 6, 9, 12, 15\}.$$

Moreover, $[n]_{18} \mapsto [0]_3$ when $n \equiv 0 \pmod{3}$. This means $n = 0, 3, 6, 9, \dots$. That is, $n \in \langle [3]_{18} \rangle$. So, $\ker \varphi = \langle [3]_{18} \rangle$. Then by the First Isomorphism Theorem, $\mathbb{Z}_{18}/\langle [3]_{18} \rangle \cong \mathbb{Z}_3$.

An alternate proof: Since $\mathbb{Z}_{18}$ is abelian, $\langle [3]_{18} \rangle$ is a normal subgroup. Moreover, it has 7 elements. Then, by Lagrange's Theorem,

$$|\mathbb{Z}_{18}/\langle [3]_{18} \rangle| = \frac{|\mathbb{Z}_{18}|}{|\langle [3]_{18} \rangle|} = \frac{18}{7} = 3.$$

Furthermore, $\mathbb{Z}_{18}$ is cyclic, so the factor group $\mathbb{Z}_{18}/\langle [3]_{18} \rangle$ is cyclic. So it is a cyclic group of order 3 and must be isomorphic to $\mathbb{Z}_3$.

Extra problems

(I) Let G_1 and G_2 be groups with $H_1 \triangleleft G_1$ and $H_2 \triangleleft G_2$. Prove the following:

(a) $H_1 \times H_2 \triangleleft G_1 \times G_2$,

(b) $(G_1 \times G_2)/(H_1 \times H_2) \cong (G_1/H_1) \times (G_2/H_2)$.

(a) We will use the Normality Test. Take $(g_1, g_2) \in G_1 \times G_2$ and $(h_1, h_2) \in H_1 \times H_2$. Then

$$(g_1, g_2)(h_1, h_2)(g_1, g_2)^{-1} = (g_1 h_1 g_1^{-1}, g_2 h_2 g_2^{-1}).$$

Since $H_1 \triangleleft G_1$, we have $g_1 h_1 g_1^{-1} \in H_1$. Similarly, $g_2 h_2 g_2^{-1} \in H_2$. Thus $(g_1, g_2)(H_1 \times H_2)(g_1, g_2)^{-1} \subseteq H_1 \times H_2$ and so $H_1 \times H_2 \triangleleft G_1 \times G_2$.

(b) Consider the map $\varphi : G_1 \times G_2 \rightarrow (G_1/H_1) \times (G_2/H_2)$ defined by $\varphi(g_1, g_2) =$

(g_1H_1, g_2H_2) . Notice that

$$\begin{aligned}\varphi((g_1, g_2)(g'_1, g'_2)) &= \varphi(g_1g'_1, g_2g'_2) \\ &= (g_1g'_1H_1, g_2g'_2H_2) \\ &= (g_1Hg'_1H, g_2Hg'_2H_2) \\ &= (g_1H_1, g_2H_2)(g'_1H_1, g'_2H_2) \\ &= \varphi(g_1, g_2)\varphi(g'_1, g'_2).\end{aligned}$$

Thus φ is a homomorphism. Moreover, it is clearly surjective. Now, $(g_1, g_2) \in \ker \varphi$ if and only if $\varphi(g_1, g_2) = (H_1, H_2)$. That is, $g_1H_1 = H_1$ and $g_2H_2 = H_2$. Meaning that $g_1 \in H_1$ and $g_2 \in H_2$. Thus $(g_1, g_2) \in \ker \varphi$ if and only if $(g_1, g_2) \in H_1 \times H_2$. By the First Isomorphism Theorem, $(G_1 \times G_2)/(H_1 \times H_2) \cong (G_1/H_1) \times (G_2/H_2)$.